

Gabriela Shamblin

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Research Interests

- Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)
- Human-Computer Interaction (HCI) and User Experience (UX)
- Medical Simulation and Education

Education

M.S. Computer Science

University of Central Florida

May 2025
Orlando, FL, USA

- Thesis: *The Impact of Visualization Styles on Movement Imitation Accuracy in Virtual Reality*
- Advisor: Dr. Joseph J. LaViola
- Mixed Reality Graduate Certificate
- GPA: 3.6/4.0
- Relevant coursework: Mixed Reality Project, Advanced Virtual Reality, AR and VR in Healthcare

B.S. Computer Science

University of Central Florida

Dec 2023
Orlando, FL, USA

- GPA: 3.7/4.0
- Relevant coursework: Artificial Intelligence, Machine Learning, Virtual Reality Engineering, Augmented Reality Engineering, Realistic Real-Time Rendering

Research Experience

Graduate Research Affiliate

Extended Reality and Artificial Intelligence (XRAI) Lab

Aug 2024 - Present
Virginia Tech

- Participate in weekly lab meetings and presenting updates on independent research under Dr. Ryan McMahan.

Undergraduate Research Assistant

Extended Reality and Training (XRT) Lab

Jan 2022 - Jul 2024
University of Central Florida

- Participated in weekly lab meetings directed by Dr. Ryan McMahan.
- Collaborated with other students on VR-based training prototypes.
- Reported on course-related technical progress and gained exposure to academic research progress.

Publications

1. **Gabriela R. Shamblin**, Ryan P. McMahan, and Joseph J. LaViola. (2026). "Investigating Gestures, Animations, and Spatial References for Acquiring Motor Skills in Virtual Reality". Submission to *ACM Conference on Human Factors in Computing Systems (CHI)*.

Professional Experience

Programmer Analyst

Johns Hopkins University

Apr 2024 - Present
Remote

- Develop immersive virtual reality training simulations for the da Vinci surgical system using Unity, C#, and Meta XR SDK to enhance user motor skills and procedural accuracy.
- Integrate advanced datasets, such as JIGSAWS, to create dynamic, interactive learning environments tailored to individual user needs, driving innovation in surgical education.
- Collaborate cross-functionally to design and implement scalable solutions, ensuring seamless functionality and user engagement for a professional audience.

Software Engineering Intern

JPMorgan Chase & Co.

Jun 2023 - Aug 2023

Tampa, FL, USA

- Designed and deployed an ADA-compliant user interface for an internal application using React.js and Springboot, supporting 30,000 users and enhancing accessibility.
- Communicated with shareholders about project specifications and collaborated with a team of six using Agile and Kanban development strategies.

Full Stack Intern

Epic North LLC

Jun 2022 - Apr 2023

Remote

- Optimized website functionality and repaired code by leveraging Angular, TypeScript, C#, ASP.NET, and SQL, ensuring seamless client deliverables.

Teaching Experience

Teaching Assistant

Concepts in Computer Science

Jul 2021 - Dec 2022

University of Central Florida

- Led twice-weekly lab sessions for 20-30 students at a time, providing hands-on guidance in foundational topics such as variables, conditionals, loops, and functions.
- Hosted weekly virtual office hours to support student learning outside of class time and answer code-related questions.
- Graded assignments for about 80 students per semester, delivering timely and constructive feedback to reinforce core programming skills.

Projects

VR Escape Room | Unity, C#, Blender

Jan 2024 - Apr 2024

- Led a team of five students to design and develop an interactive puzzle-solving experience in Unity, integrating custom assets and lighting effects.

SenseRator | Python, PySimpleGUI, Open3D

Feb 2023 - Dec 2023

- A Python-based application integrating lidar and camera technology to perform object detection and semantic segmentation.

Virtual Reality Engineering | Unity, C#, SteamVR

Jan 2023 - Apr 2023

- A series of six Unity projects covering topics of environments, traveling, physics, animations, and user interactions.

Technologies

Languages: C#, Python, R, Java, JavaScript

Libraries: Pandas, NumPy, Matplotlib, Sklearn, TensorFlow

Engines & Toolkits: Unity, XR Interaction Toolkit, SteamVR, Meta XR SDK, Unreal Engine

XR Hardware: Meta Quest 2, Meta Quest 3, HTC Vive

Tools: Git